

Roger Tawfik

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Education

University of Massachusetts Amherst

Amherst, MA

Candidate for Bachelor of Science in Computer Engineering

Expected May 2028

- **Academic Distinctions:** GPA: 3.804/4.00 | Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- **Relevant Coursework:** Advanced Programming, Embedded Systems, Circuits & Electronics, Digital & Computer Systems

Technical Skills

Languages: Python, TypeScript, C, C++, Java, Assembly (ARM/AVR)

Full-Stack: React, Vite/PWA, Supabase (Auth/Postgres/Edge Functions), REST API integration

Core CS: Data Structures, Algorithms, Object-Oriented Programming

ML, AI & Computer Vision: LangChain, YOLOv8, MediaPipe, OpenCV, TensorFlow, MeTRAbs, XGBoost, Random Forest

Data & Tools: Git, Linux, Pandas, NumPy, SQLite, LM Studio, GCC, MATLAB

Embedded Systems: Arduino firmware development, Concurrency, Digital Logic, Circuits & Electronics

Work Experience

I-SENSE, Florida Atlantic University

Jun 2026 - Jul 2026

NSF REU Researcher (Undergraduate Research Intern)

Boca Raton, FL

Python, OpenCV, MediaPipe, YOLOv8, MeTRAbs, LM Studio, Git

Health & Behavior: Wearables & Data Analytics for Personalized Therapeutic Management, advised by Dr. Behnaz Ghoraani
[\[Project Page\]](#)

- Developed a clinical gait video analysis pipeline to evaluate walking recordings for quality, extract body-joint data, and support automated gait-metric computation
- Built a three-tier quality-control system screening recordings for lighting, focus, subject visibility, and walking orientation before downstream pose estimation
- Integrated a local LLM via LangChain to generate dual-audience QC reports (technical summaries for developers, plain-language guidance for clinicians)

Personal Projects

AI Nutrition Tracker App | TypeScript, React, Supabase, Vite/PWA

Jul 2026 - Present

- Built a nutrition tracker for natural language, photo, and barcode food logging, with adaptive calorie targets, gap-fill nutrient recommendations, and daily AI coaching insights
- Architected as a mobile-first PWA with Supabase auth, Postgres, and edge functions, using a review-before-log workflow to keep AI-parsed entries accurate

Gym Form Trainer | Python, C, OpenCV, MediaPipe, Arduino

Dec 2025 - May 2026

- Built a real-time computer vision fitness trainer in Python using MediaPipe Pose and OpenCV for full-body landmark extraction, HUD rendering, and color-coded form scoring
- Designed and flashed C firmware to an Arduino microcontroller for live LED/buzzer feedback via bidirectional serial communication, closing the loop from sensor input to physical hardware output

Algorithmic Trading Analysis System | Python, Pine Script, SQLite, XGBoost

Feb 2026 - Apr 2026

- Engineered an end-to-end automated market analysis system in Python to screen 5,000+ stocks using multi-factor technical signals and quantitative scoring models for breakout detection
- Built and processed 150,000+ labeled observations to train and calibrate XGBoost models into a probabilistic trade-quality scoring framework, with SQLite caching and TradingView Pine Script backtesting

Service Mesh Graph Project | Python

Nov 2025 - Dec 2025

- Designed and implemented a weighted graph-based service mesh simulation in Python using OOP and modularity
- Built BFS reachability, Dijkstra shortest path, and Prim MST analyses for connectivity and optimization